

MICROCONTROLLER AND APPLICATIONS

Sr.No	QUESTIONS	Reference
1	Explain the term Programmable device.	MICRO pdf
2	Explain the difference between Microprocessor and Microcontroller.	MICRO pdf
3	Discuss the difference between how execution carried out in a Computer and in a Microcontroller.	MICRO pdf
4	Give details of some parameters affecting the speed of code execution for a microcontroller.	MICRO pdf
5	Explain hoe pipelining and Cache memory reduces speed of execution.	MICRO pdf
6	Explain Von numen Architecture and their applications.	MICRO pdf
7	Explain difference between Von numen and Harvard Architecture, with required diagram.	MICRO pdf
8	Explain Modified Harvard architecture.	MICRO pdf
9	Classify the microcontroller 8051 and ARM in to Von numen and/or Harvard architecture with required justification and reason for the classification answered.	MICRO pdf
10	List and explain the criteria for selection of microcontroller, for implementation of desired application/system.	MICRO pdf
11	Explain various Peripherals available in microcontroller.	MICRO pdf
12	Explain machine cycle & related operations	YT video
13	Explain block diagram, programming model, pin diagram & internal ram organisation of 8051 microcontroller	reference 1
14	Explain timer counter in different modes [mode 0,1,2]	Class notes
15	Explain various registers [TMOD, TCON & IE]	Class notes
16	Explain interrupt vs polling	Class notes
17	Explain interrupt vector (google)	google
18	Program for timer, interfacing the key, driving an LED	Class notes
19	Assembly program for Arithmetic, Logical, MOV	reference 2
20	Role of programmable device in control system	Class notes
21	What is microcontroller? Definition of microcontroller	Class notes
22	Various components of microcontroller.	Class notes
23	Different peripherals in microcontroller.	Class notes
24	Pin & block diagram of 8051 microcontroller	reference 1
25	Explain how microcontroller executes the code	Class notes
26	Difference between assembly & C programming	Class notes
27	Basic structure of c & asm	Class notes
28	Assembly programming -Addressing mode & instruction [MOV] -Arithmetic instruction [Operation] -logical instruction -Stack related instruction [memory]	reference 2

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29	C programming -difference between C-program & embedded [structure of embedded C-programming] -Basic concept in C-programming [Loops, Variable declaration, user defined function] -Key words [Special key words] -Port & port pin as O/P port & O/P pin & its C program -Port & port pin configured as I/P pin -Key bouncing & de-bouncing circuit	Class notes
30	Explain asm directives	reference 2
31	Explain terms -Types of resistors, accumulator, program counter & data pointer , instruction decoder, stack & stack pointer, special function resistor, flag and program status word	reference 1
32	Timer calculations	Class notes

Unit No.	Topics	Teaching Hours
1	Microcontroller Basics: Difference between microprocessor and microcontroller, CISC vs RISC design philosophy, Von Neumann vs Harvard architecture. Introduction to MCS -51 Family microcontrollers, address bus, data bus, control signals, clock and reset circuits, working registers, special function registers, stack and use of stack pointer, program counter, I/O Ports, power saving modes and its operation, timing diagram for execution cycle.	05
2	8051 Microcontroller Architecture: 80C51 Central Processing Unit Diagram, Architectural block diagram, Pin diagram and Pin Functions, memory organization, Internal program and data memory	07
3	On Chip Peripherals: Timer/Counters and associated registers, Various modes of timer/counter operations, Concept of Interrupt, Interrupt versus polling, Types of Interrupts in 8051, Interrupt control and associated registers, Interrupt vectors, Interrupt execution, nested interrupts, Basics of serial communication, Serial data input/output and associated registers, Various modes of serial data communication.	08